

Vector Field Approach for Curved Path Following for Miniature Aerial Vehicles

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A robust path following method that allows miniature aerial vehicles to track straight lines, arcs, and arbitrary curved paths using a vector field of course commands is presented. In addition, a proof showing that lateral-following error decays asymptotically even in the presence of constant wind disturbances is put forth. These claims are experimentally verified with flight test results that demonstrate accurate path following over a variety of curved paths with varying spatial frequency and amplitude. The average lateral-following error in preliminary results is 3.4 m for amplitudes ranging between 10 and 100 m and spatial periods between 125 and 500 m.

I. Introduction

For most applications of unmanned aerial vehicles (UAVs), the ability to follow paths composed of straight lines and arcs is sufficient to accomplish mission objectives. However, certain applications require accurate following of curved paths including tactical maneuvers, formation flying, convoy support,¹ and low altitude missions through narrow mountainous canyons.

There are several advantages associated with using miniature aerial vehicles^a (MAVs) to accomplish many of these missions. In addition to being less costly than larger UAVs, they are also more easily transported and can be launched even in rough terrain. In general, they are also less complex and therefore require fewer ground operators. Their simplicity also allows MAVs to be deployed more quickly than larger UAV systems. Possibly, the greatest advantage inherent in MAVs is their heightened potential to navigate narrow passages and confined urban environments. For fixed-wing MAVs, the minimum turn radius is typically in the range of 10 to 50 m compared to the range of 250 to 2000 m for large non-aerobatic UAVs like Predator and Global Hawk. This translates into a significant increase in the spatial frequency of paths that can be tracked by MAVs. Path tracking algorithms must be able to utilize the full turning capability of the MAV in order to exploit this key advantage.

Numerous approaches have been taken for curved path trajectory tracking and following. In one approach,² an optimal path is divided into segments. Each segment is then approximated using a 7th-order polynomial. These polynomial path segments are then followed using a controller derived from a kinematic model with the addition of an anticipatory control element that improves the tracking capability for curved paths. This method also accounts for wind disturbances by the addition of an adaptive kinematic factor. An experimental tracking error of 1.6 m RMS is reported. Accurate curved path following has also been achieved in simulation for autonomous underwater vehicles (AUVs) by Lapierre, et al.³ While the dynamics of AUVs are notably different than those of UAVs, the Lyapunov approach taken by the authors to derive and prove a nonlinear path following control law is insightful for similar path following approaches for UAVs. A vector pursuit method of path following for unmanned ground vehicles generates a desired orientation based on the current position and orientation of the vehicle relative to the position of a look-ahead point on the planned path and the desired orientation along the path at the look-ahead point. Experimental results show that this technique is less sensitive to the look-ahead distance and vehicle speed while still providing an anticipatory element to improve tracking on paths with high curvature.⁴ A new methodology for the design of UAV guidance, navigation, and control systems attempts to design the guidance and control simultaneously. This integrated approach achieves zero steady-state error at any trimming trajectory by using gain scheduling theory. This approach also addresses the stability of the combined guidance and control system.⁵

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^aIn this paper miniature aerial vehicles are considered to be those with wingspans in the 0.3 m to 2 m range and micro aerial vehicles to be those with wingspans under 0.3 m. Here the abbreviation MAV denotes *miniature* aerial vehicle.

An important limitation of MAVs is that they are much more adversely affected by windy conditions than large UAVs. This limitation is mostly caused by their light weight and the low range of airspeeds of which they are capable. Since wind speeds can often surpass 40 percent of the MAV airspeed, effective path tracking strategies must be developed to overcome the effects of these environmental disturbances.

One successful and robust method that allows MAVs to follow straight lines and arcs in the presence of wind has been developed by Nelson, et al.⁶ This path following approach uses a vector field of course commands to guide the MAV on the waypoint path with minimal overshoot. Experimental results have yielded cross-tracking errors of 0.7 m on average with wind speeds up to 50 percent of the MAVs airspeed.

The work presented in this paper builds on the work presented by Nelson, et al. by adapting their vector field approach for straight line paths and arcs to extend to arbitrary curved paths. One advantage of this method is that the vector field approach lends itself well to reactive collision avoidance methods that alter the path in real time as obstacles are detected.⁷ In addition, this vector field approach takes into account the instantaneous speed of the MAV which allows for close following with minimal overshoot over a wide range of groundspeeds. Experimental results validate the feasibility of this approach for MAVs and demonstrate accurate curved path following even in moderately windy conditions.

II. Problem Description

The objective of this paper is to present a new method for MAVs to accurately follow curved paths in the presence of wind. This method is designed to work for all curved paths including piecewise continuous functions composed of straight lines, arcs, and curves without requiring the addition of any special cases for each path type. This method will also work on any arbitrary curve that can be divided into a series of path segments based on angular and spatial constraints. Each segment can then be approximated with polynomial path segments using linear regression.²

Unlike trajectory tracking where the objective is to track a moving point in time, the objective of path following is to spatially track a path without regard for temporal constraints.⁸ The path following approach in this paper uses a vector field around the path to be followed to guide the MAV onto the path asymptotically. Figure 1 shows an example of a vector field for a curved path segment.

III. Technical Approach

For this method, it is assumed that altitude h is accurately controlled by pitch and altitude control loops. It is also assumed that the airspeed of the fixed-wing MAV V_a is maintained at a safe level above stall speed V_{stall} . A model of the navigational dynamics used for this path following approach are given according to

$$\dot{x} = V_g \cos \chi \quad (1)$$

$$\dot{y} = V_g \sin \chi, \quad (2)$$

where V_g is the groundspeed and χ is the course of the MAV. In using these navigational dynamics, constant wind disturbances are already accounted for since the motion of the MAV is represented in terms of groundspeed and course which are independent of the wind speed and direction. Another key assumption is that the MAV follows first-order course dynamics given by

$$\dot{\chi} = \alpha (\chi^c - \chi), \quad (3)$$

where χ^c is the commanded course, and α is a positive constant that physically represents the reciprocal of the time constant.

A. Control Approach

Let d be the lateral distance from the curved path to the MAV as shown in Figure 2. Another important variable is the path course χ^p . For the curve, $y = f(x)$, χ^p can be found according to

$$\chi^p = \tan^{-1} \left(\frac{dy}{dx} \right). \quad (4)$$

The rate of change of the lateral-tracking error $\dot{d}$ is defined as a function of the groundspeed V_g and relative orientation by

$$\dot{d} = V_g \sin(\chi - \chi^p). \quad (5)$$

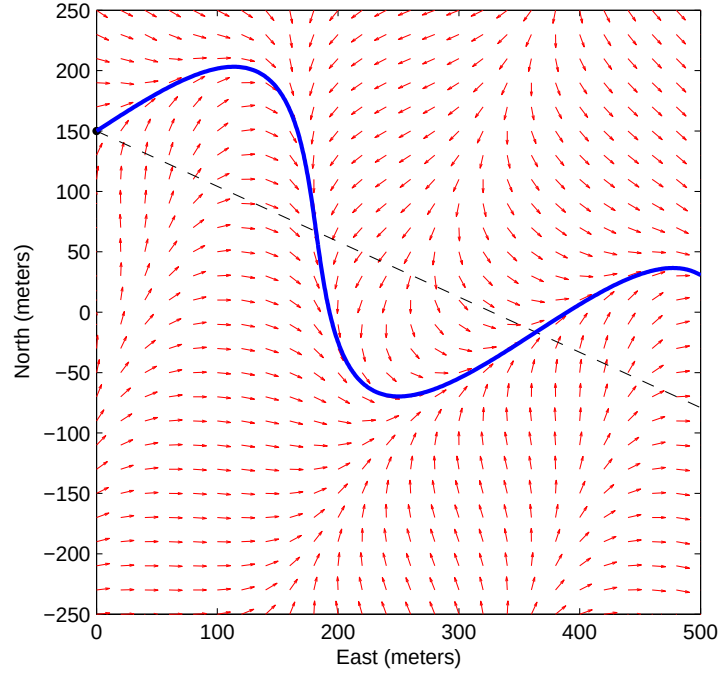


Figure 1. Curved path vector field. The solid blue line represents the curve path to be followed and the dashed black line indicates the directional sense of the curve.

To direct the MAV onto the path, an asymptotically stable vector field of course commands is constructed around the curved path segment according to

$$\chi^d(d) = \chi^p - \chi^\infty \frac{2}{\pi} \tan^{-1}(kd) \quad (6)$$

where $\chi^d(d)$ is the desired course expressed as a function of the lateral-tracking error d and χ^p is the course of the path at the point from which d was computed. Note that as d gets smaller, the desired course transitions from χ^∞ at $d = \infty$ to χ^p at $d = 0$. The positive constant k controls the rate of this transition and can be thought of as the strength of the vector field. Ultimately, for the MAV to get on the path this approach must drive d to zero by driving χ toward χ^d . The angle χ^∞ is limited to the range $(0, \frac{\pi}{2}]$ so that

$$-\frac{\pi}{2} < \chi^\infty \frac{2}{\pi} \tan^{-1}(kd) < \frac{\pi}{2}$$

over all values of d .

To solve for the appropriate control signal χ^c that will force the tracking error to converge, first consider the Lyapunov function $W_1 = \frac{1}{2}d^2$ with derivative

$$\begin{aligned} \dot{W}_1 &= d\dot{d} \\ &= dV_g \sin(\chi - \chi^p) \end{aligned} \quad (7)$$

so that if $\chi = \chi^d$, then $d \rightarrow 0$ asymptotically. In this paper a sliding mode approach will be used to ensure that $\chi = \chi^d$. Under the assumption that $\chi = \chi^d$ and substituting (6) into (7), we find that

$$\dot{W}_1 = -dV_g \sin\left(\chi^\infty \frac{2}{\pi} \tan^{-1}(kd)\right), \quad (8)$$

which is negative definite thus meeting Lyapunov stability criteria and ensuring $d \rightarrow 0$ asymptotically. Next we must

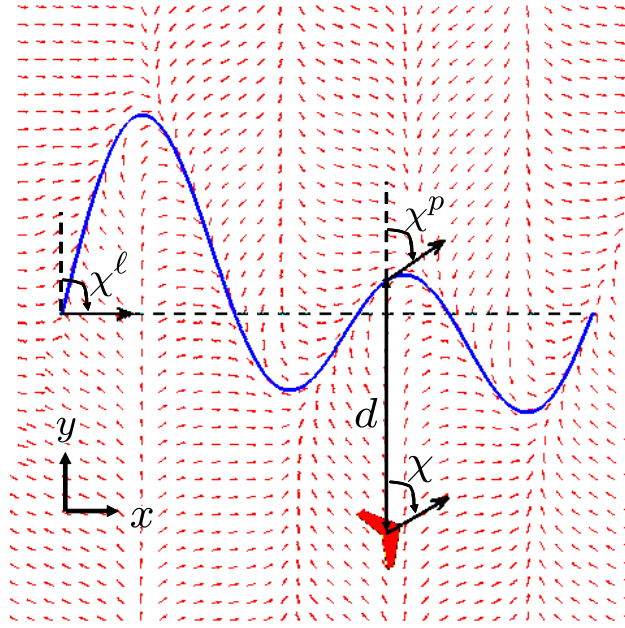


Figure 2. Vector field for curved path following.

derive a controller to ensure $\chi \rightarrow \chi^d$. Let $\tilde{\chi} \triangleq \chi - \chi^d(d)$ and differentiate to give

$$\begin{aligned}\dot{\tilde{\chi}} &= \dot{\chi} - \dot{\chi}^d(d) \\ &= \alpha(\chi^c - \chi) - \dot{\chi}^p + \frac{2\chi^\infty}{\pi} \frac{k}{1 + (kd)^2} V_g \sin(\chi - \chi^p).\end{aligned}\quad (9)$$

Next, let $W_2 = \frac{1}{2}\tilde{\chi}^2$. Differentiating and substituting from (9) gives

$$\begin{aligned}\dot{W}_2 &= \tilde{\chi}\dot{\tilde{\chi}} \\ &= \tilde{\chi} \left(\alpha(\chi^c - \chi) - \dot{\chi}^p + \frac{2\chi^\infty}{\pi} \frac{k}{1 + (kd)^2} V_g \sin(\chi - \chi^p) \right).\end{aligned}\quad (10)$$

To ensure that (10) is negative definite we will choose the control command as

$$\chi^c = \chi + \frac{\dot{\chi}^p}{\alpha} - \frac{1}{\alpha} \frac{2\chi^\infty}{\pi} \frac{k}{1 + (kd)^2} V_g \sin(\chi - \chi^p) - \frac{\beta}{\alpha} \text{sign}(\tilde{\chi}).\quad (11)$$

where β is a positive constant and

$$\text{sign}(\tilde{\chi}) = \begin{cases} 1 & \text{if } \tilde{\chi} > 0 \\ 0 & \text{if } \tilde{\chi} = 0 \\ -1 & \text{if } \tilde{\chi} < 0. \end{cases}$$

As a result, $\dot{W}_2 \leq -\beta|\tilde{\chi}|$ and we can conclude that $\tilde{\chi} \rightarrow 0$ in finite time.⁹ Unfortunately, using $\text{sign}(\tilde{\chi})$ causes undesirable control chatter in implementation. One solution to this problem is to replace $\text{sign}(\tilde{\chi})$ with $\text{sat}(\frac{\tilde{\chi}}{\epsilon})$ where

$$\text{sat}\left(\frac{\tilde{\chi}}{\epsilon}\right) = \begin{cases} \frac{\tilde{\chi}}{\epsilon} & \text{if } \left|\frac{\tilde{\chi}}{\epsilon}\right| \leq 1 \\ \text{sign}\left(\frac{\tilde{\chi}}{\epsilon}\right) & \text{otherwise} \end{cases}.$$

The width of the sliding mode boundary region is represented by the positive constant ϵ . Implementing this change results in proportional feedback around the course error inside the boundary region and yields the final control command

$$\chi^c = \chi + \frac{\dot{\chi}^p}{\alpha} - \frac{1}{\alpha} \frac{2\chi^\infty}{\pi} \frac{k}{1 + (kd)^2} V_g \sin(\chi - \chi^p) - \frac{\beta}{\alpha} \text{sat}\left(\frac{\tilde{\chi}}{\epsilon}\right).\quad (12)$$

It is important to note that large values of k will cause large changes in χ^d as the MAV gets close to the curved path that are difficult to track. Equally important is the term $\dot{\chi}^p$ which represents the rate of change of the path course over time, which will be discussed in some detail in Section B.

Theorem III.1 *The system of equations given by (1), (2) and (9), where χ^c is given by (12) and $\chi^d(d)$ is given by (6) is asymptotically stable when $0 < \chi^\infty \leq \frac{\pi}{2}$ and*

$$\left(\frac{2\chi^\infty k}{\pi\epsilon\mu}\right) \min \left\{ \left(\frac{\rho\beta}{V_g}\right), \left(\frac{\bar{d}}{2}\right) \right\} > 1, \quad (13)$$

where $\bar{d}$ is arbitrarily chosen and

$$\mu \geq \max \left\{ \frac{2(1 + (k\bar{d})^2)}{\cos\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)}, \frac{4\chi^\infty k}{\pi \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)} \right\}.$$

The proof of Theorem III.1 is included in the Appendix. By observation, it will be difficult to satisfy (13) for large values of V_g . Intuitively this makes sense. Higher groundspeeds result in a larger turning radius which will make a curved path more difficult or impossible to track. Small values of ϵ will enhance tracking by increasing the gain. Small values of β and k will make it difficult to satisfy this constraint. However, large values of β could cause unrealistic control effort. In summary, the control command in (12) drives χ to $\chi^d(d)$ in finite time using the course field described by (6). In turn, this will lead to asymptotic decay of lateral-tracking error as long as (13) is satisfied.

B. Implementation Issues

The physical heading rate constraints associated with fixed-wing MAVs place a limit on the path curvature that can accurately be followed. The path course rate $\dot{\chi}^p$, represented in the control command in (12), is an important factor in determining whether a curved path is flyable. It also has physical meaning, representing the functional relationship between the groundspeed and relative orientation of the MAV and the spatial rate of change of the path. If the curved path to be followed is C^2 continuous, then $\dot{\chi}^p$ can be solved for explicitly. Consider the sinusoidal path

$$y = A \sin(\omega x), \quad (14)$$

where A is the amplitude, ω is the spatial frequency measured in radians per meter, and x is the distance along the path in meters. The path course rate $\dot{\chi}^p$ can be found by first expressing it as

$$\dot{\chi}^p = \frac{d\chi^p}{dx} \frac{dx}{dt}, \quad (15)$$

which allows the solution to be divided into two parts. The first part, $\frac{d\chi^p}{dx}$, is found by taking the expression for χ^p given by

$$\begin{aligned} \chi^p &= \tan^{-1} \left(\frac{dy}{dx} \right) \\ &= \tan^{-1}(A\omega \cos(\omega x)), \end{aligned} \quad (16)$$

and differentiating to yield

$$\frac{d\chi^p}{dx} = \frac{-A\omega^2 \sin(\omega x)}{1 + A^2\omega^2 \cos^2(\omega x)}. \quad (17)$$

The second part, $\frac{dx}{dt}$, is related to the groundspeed of the MAV according to

$$\frac{dx}{dt} = V_g \cos(\chi^\ell - \chi), \quad (18)$$

where V_g is groundspeed and χ^ℓ is the directional sense of the path segment as shown in Figure 2. Finally, the solution is obtained by substituting (17) and (18) into (15) to yield

$$\dot{\chi}^p = \frac{-A\omega^2 \sin(\omega x) V_g \cos(\chi^\ell - \chi)}{1 + A^2\omega^2 \cos^2(\omega x)}. \quad (19)$$

The maximum value for $\dot{\chi}^p$ occurs at the peaks of the sine wave or when $x = \frac{n\pi}{2}$, where n is an odd integer. At its maximum value, (19) can be reduced to

$$\dot{\chi}_{max}^p = |A\omega^2 V_g|. \quad (20)$$

In Figure 3, a contour plot shows how $\dot{\chi}_{max}^p$ varies with spatial frequency and amplitude for sinusoidal paths while holding a constant groundspeed. The spatial frequency has the greatest effect in determining whether or not a path is flyable since $\dot{\chi}_{max}^p \propto \omega^2$. For straight line segments, $\frac{d\chi^p}{dx} = 0$ and therefore $\dot{\chi}^p$ will have a value of zero, thus simpli-

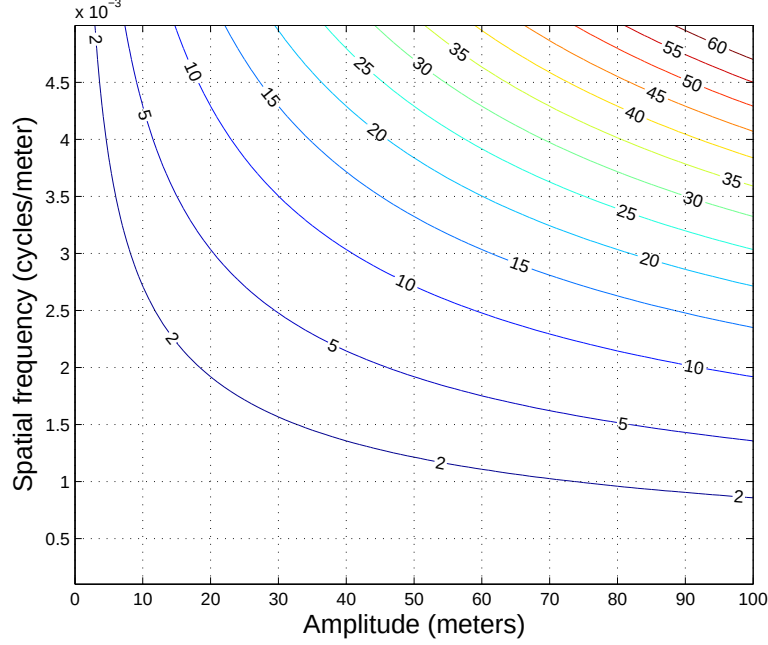


Figure 3. Contour plot of $\dot{\chi}_{max}^p$ in deg/s for a constant groundspeed $V_g = 13$ m/s varying spatial frequency and amplitude for a sinusoidal path.

fying the problem. When following circular paths, $\dot{\chi}^p$ will be held relatively constant only varying with groundspeed. For curved paths, if $\dot{\chi}^p$ is ever greater than the maximum achievable turn rate of the MAV, the MAV will not be able to closely follow the curve until $\dot{\chi}^p$ drops to a manageable level. One way to reduce $\dot{\chi}_{max}^p$ without changing the path is to reduce the MAV groundspeed by commanding a lower airspeed. This is only effective to a certain degree and caution must be taken to keep the MAV airspeed above V_{stall} . For path segments represented by polynomials, $\dot{\chi}^p$ can also be found in general terms following the same steps outlined in (15) – (19).

In practice, $\dot{\chi}^p$ can be approximated using a simple backward difference numerical derivative according to

$$\dot{\chi}^p \approx \frac{\chi^p - \chi_{old}^p}{dt}, \quad (21)$$

where χ_{old}^p is the path course computed in the previous time step. In the limit, as dt approaches zero, (21) approaches the true value of $\dot{\chi}^p$. This numerical method has the advantage of being both computationally cheap and effective at estimating $\dot{\chi}^p$ for any path type, not exclusively paths represented by known differentiable functions.

C. Effects of Model Error

Up to this point it has been assumed that the constant α in the course rate model in (3) was accurately known. However, α is difficult to determine experimentally and inevitably there will be some error in its estimated value, which will result in less precise path following. In this section, limits will be placed on parameters as a function of the percent error in α . First, let $\alpha = \hat{\alpha} + \tilde{\alpha}$ where $\hat{\alpha}$ is the estimate of α and $\tilde{\alpha}$ is the error in $\hat{\alpha}$. The course rate model can then be expressed according to

$$\dot{\chi} = \hat{\alpha}(\chi^c - \chi) + \tilde{\alpha}(\chi^c - \chi). \quad (22)$$

Recalling that

$$\begin{aligned}\tilde{\chi} &= \chi - \chi^d \\ &= \chi - \chi^p + \frac{2\chi^\infty}{\pi} \tan^{-1}(kd),\end{aligned}$$

its derivative becomes

$$\begin{aligned}\dot{\tilde{\chi}} &= \dot{\chi} - \dot{\chi}^p + \frac{2\chi^\infty}{\pi} \frac{k\dot{d}}{1 + (kd)^2} \\ &= -\hat{\alpha}\chi + \hat{\alpha}\chi^c + \tilde{\alpha}(\chi^c - \chi) - \dot{\chi}^p + \frac{2\chi^\infty}{\pi} \frac{k\dot{d}}{1 + (kd)^2}.\end{aligned}\quad (23)$$

Then taking the control signal in (12) and substituting it into (23) yields

$$\dot{\tilde{\chi}} = -\beta \text{sign}(\tilde{\chi}) + \left(\frac{\tilde{\alpha}}{\hat{\alpha}} \dot{\chi}^p - \frac{\tilde{\alpha}}{\hat{\alpha}} \frac{2\chi^\infty}{\pi} \frac{k\dot{d}}{1 + (kd)^2} - \frac{\tilde{\alpha}}{\hat{\alpha}} \beta \text{sign}(\tilde{\chi}) \right). \quad (24)$$

Assuming that $\hat{\alpha}$ is in error by less than 100 percent or

$$\left| \frac{\tilde{\alpha}}{\hat{\alpha}} \right| < k_o < 1, \quad (25)$$

then using the Lyapunov function $W_1 = \frac{1}{2} \tilde{\chi}^2$ and differentiating gives

$$\begin{aligned}\dot{W}_1 &= \tilde{\chi} \dot{\tilde{\chi}} \\ &= \tilde{\chi} \left(-\beta \text{sign}(\tilde{\chi}) + \frac{\tilde{\alpha}}{\hat{\alpha}} |\dot{\chi}^p| - \frac{\tilde{\alpha}}{\hat{\alpha}} \frac{2\chi^\infty}{\pi} kV_g - \frac{\tilde{\alpha}}{\hat{\alpha}} \beta \text{sign}(\tilde{\chi}) \right) \\ &< -\beta |\tilde{\chi}| + \frac{\tilde{\alpha}}{\hat{\alpha}} |\dot{\chi}^p| |\tilde{\chi}| + \left| \frac{\tilde{\alpha}}{\hat{\alpha}} \frac{2\chi^\infty}{\pi} kV_g \right| |\tilde{\chi}| - \frac{\tilde{\alpha}}{\hat{\alpha}} \beta |\tilde{\chi}| \\ &< -|\tilde{\chi}| \left(-k_o |\dot{\chi}^p| - k_o \frac{2\chi^\infty}{\pi} kV_g + \beta(1 + k_o) \right).\end{aligned}\quad (26)$$

To satisfy the Lyapunov criteria and give finite-time convergence to the sliding surface, $\dot{W}_1$ must be less than zero. This will be true if

$$\beta \geq \frac{k_o \left(\frac{2\chi^\infty}{\pi} kV_g + |\dot{\chi}^p| \right)}{1 + k_o}. \quad (27)$$

When $\tilde{\alpha}$ is small, k_o will be small and (27) can be satisfied more easily. By observation, since $|\dot{\chi}^p|$ is a function of groundspeed (27) will be difficult to satisfy for paths with sharp curvature at high groundspeeds.

D. Bounding Following Error

Everything above has ignored error caused by course-rate saturation, unmodeled dynamics, and imperfect course measurements, all of which will degrade path following performance. Much of this error will emerge in $\tilde{\chi}$ preventing it from decaying to zero. In this section, a method to show input-to-state stability (ISS) can be used to bound the lateral following error as a function of $\tilde{\chi}$.⁹ In the process of showing that (5) is ISS, we will find an equation $\rho(|\tilde{\chi}|)$ that will bound lateral-tracking error so that

$$\dot{V} < -W(d) \quad \forall \quad d > \rho(|\tilde{\chi}|), \quad (28)$$

where $W(d)$ is continuous and positive definite, and $\rho(|\tilde{\chi}|)$ is a class $\mathcal{K}$ function.⁹ Letting $V = \frac{1}{2} d^2$ then

$$\begin{aligned}\dot{V} &= d\dot{d} \\ &= dV_g \sin(\chi - \chi^p) \\ &= dV_g \sin(\chi^d(d) + \tilde{\chi} - \chi^p) \\ &= dV_g \sin(\tilde{\chi}^d(d) + \tilde{\chi}).\end{aligned}\quad (29)$$

To continue we will define a constant $\theta \in (0, 1)$ and rewrite (29) as

$$\dot{V} = dV_g \sin((1 - \theta)\tilde{\chi}^d(d) + \theta\tilde{\chi}^d(d) + \tilde{\chi}). \quad (30)$$

Noticing that $dV_g \sin((1 - \theta)\tilde{\chi}^d(d)) > 0$ we will let $W(d) = -dV_g \sin((1 - \theta)\tilde{\chi}^d(d))$. Next, conditions on $|d|$ must be found such that

$$dV_g \sin((1 - \theta)\tilde{\chi}^d(d) + \theta\tilde{\chi}^d(d) + \tilde{\chi}) < dV_g \sin((1 - \theta)\tilde{\chi}^d(d)). \quad (31)$$

It is important to note that the sine function is monotonic over the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. There are two cases that will be examined to show satisfaction of (31).

Case I: ($d > 0$) $\Leftrightarrow \tilde{\chi}(d) < 0$

$$\begin{aligned} \tilde{\chi} + \theta\tilde{\chi}^d(d) &< 0 \\ \tilde{\chi} &< -\theta\tilde{\chi}^d(d) \\ \tilde{\chi} &< \theta\chi^\infty \frac{2}{\pi} \tan^{-1}(kd) \end{aligned} \quad (32)$$

Case II: ($d < 0$) $\Leftrightarrow \tilde{\chi}(d) > 0$

$$\begin{aligned} \tilde{\chi} + \theta\tilde{\chi}^d(d) &> 0 \\ \tilde{\chi} &> -\theta\tilde{\chi}^d(d) \\ \tilde{\chi} &> \theta\chi^\infty \frac{2}{\pi} \tan^{-1}(kd) \end{aligned} \quad (33)$$

Combining (32) and (33) gives

$$|\tilde{\chi}| < \theta\chi^\infty \frac{2}{\pi} \tan^{-1}(k|d|). \quad (34)$$

Then solving for $|d|$ yields

$$|d| > \frac{1}{k} \tan\left(\frac{|\tilde{\chi}|\pi}{2\theta\chi^\infty}\right), \quad (35)$$

which function is strictly increasing and is equal to zero when the input $\tilde{\chi} = 0$, thus meeting the requirements of a class $\mathcal{K}$ function.⁹ Letting

$$\rho(|\tilde{\chi}|) = \frac{1}{k} \tan\left(\frac{|\tilde{\chi}|\pi}{2\theta\chi^\infty}\right) \quad (36)$$

satisfies (28) which can now be written as

$$\dot{V} < -dV_g \sin((1 - \theta)\tilde{\chi}^d(d)) \quad \forall \quad |d| > \frac{1}{k} \tan\left(\frac{|\tilde{\chi}|\pi}{2\theta\chi^\infty}\right). \quad (37)$$

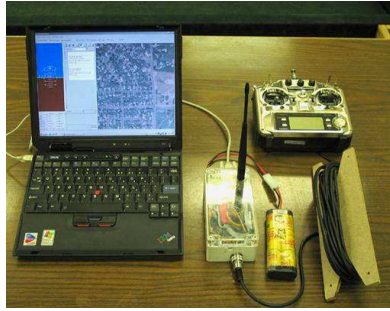
This represents a conservative bound on the lateral-tracking error. The significance of this is that the magnitude of lateral-tracking error $|d|$ is guaranteed to decrease asymptotically until it is less than $\rho(|\tilde{\chi}|)$ where it will remain thereafter. However, it cannot be shown that $|d|$ will converge to zero. For small values of $\tilde{\chi}$, $\rho(|\tilde{\chi}|)$ will be small and therefore the bound on $|d|$ will be tight, resulting in accurate curved path following as long as $\dot{\chi}^p$ does not exceed the maximum turn rate of the MAV, as discussed in Section B. In the following example, we will select typical values for the parameters in (35). For $W(d)$ to be negative definite, θ must be less than 1. However, to get the tightest bound possible, θ must be approximately 1, thus we will let $\theta = 0.99$. Letting $k = 0.02$, and $\chi^\infty = \frac{\pi}{2}$, when $\tilde{\chi} = 4$ degrees, the bound on $|d|$ can then be found as

$$\begin{aligned} |d| &> \frac{1}{0.02} \tan\left(\frac{|4\frac{\pi}{180}|\pi}{2(0.99)(\frac{\pi}{2})}\right) \\ &> 3.5 \text{ m}, \end{aligned}$$

thus the lateral-tracking error will decay asymptotically until it is less than 3.5 m.

IV. Experimental Apparatus

BYU has developed a reliable and robust platform for MAV testing. Figure 4(a) shows the ground station components. A laptop runs the Virtual Cockpit software that interfaces through a communication box with the MAV over a wireless modem. An RC transmitter is used as a stand-by fail-safe mechanism. At any time during flight tests an RC pilot can assume control of the MAV to help facilitate safe operations.



(a) MAV Ground station hardware



(b) MAV flying wing airframe



(c) The Kestrel autopilot developed at BYU

Figure 4. Main elements of MAV testbed.

Figure 4(b) shows the airframe used for most of the flight tests reported in this paper. The airframe is a Unicorn flying wing with an EPP foam core and a 0.9 m wingspan. This model was selected for its durability, payload capacity, ease of component installation, and flight characteristics. Embedded in the airframe are the autopilot, lithium-polymer batteries, a 900 MHz radio modem, a 12 channel GPS receiver, a small analog video camera, and a video transmitter.

The path following methods described in this paper were implemented on the Kestrel 2.1 autopilot made by Procerus Technologies shown in 4(c). The Kestrel autopilot is equipped with a Rabbit 3400 29 MHz processor, three-axis rate gyros, three-axis accelerometers, and absolute and differential pressure sensors. The autopilot measures $3.5 \times 5.0 \times 1.2$ cm and weighs 16.5 grams. The autopilot also serves as a data acquisition device and is able to log 175 kbytes of user-selectable telemetry at rates up to 60 Hz.

V. Flight Test Results

In flight tests, a MAV was commanded to fly a variety of sinusoidal path segments with varying amplitudes and spatial periods. During the flight tests the average wind speed measured 2.4 m/s (18 percent of the commanded airspeed). The direction of the wind was from the northeast. Wind gusts of 3.2 m/s (26 percent of the commanded airspeed) were also recorded. Plots of flight test results with accompanying RMS following error (RMSE) are shown in Figure 5. The mean following error on the path with a period of 350 m was 3.7 m with a standard deviation of 3.75 m and a $\dot{\chi}_{\max}^p$ of 35 deg/sec. As expected, the MAV was able to accurately follow the large period sinusoids with an average error of less than 3 m for the three paths where the period was 500 m. The mean error was lowest for the path shown in Figure 5(e) at only 2.5 m with a standard deviation of 3.4 m. Unsurprisingly, this path also had the lowest $\dot{\chi}_{\max}^p$ at 5.5 deg/sec. These results also show that the MAV had difficulty precisely following the path with the shortest period, $T = 125$ m, and a corresponding $\dot{\chi}_{\max}^p = 19$ deg/sec. The mean lateral-tracking error for the 125 m period path was 5.1 m, while the standard deviation of the path error was 4.3 m.

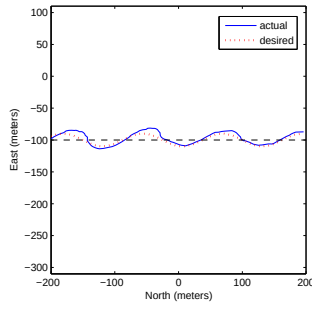
In another set of flight tests, the MAV followed two irregularly shaped circular paths in a counterclockwise direction. During these tests the average recorded wind speed was 4 m/s (31 percent of the commanded airspeed) gusting from the east-southeast. The path shown in Figure 6(a) had an average amplitude $\bar{A}$ of 38 m and its shortest spatial period $T_{\min}$ was 315 m. The mean error when following this low-frequency path was 3.5 m. The mean error increased to 3.8 m for the high-frequency path shown in Figure 6(b) where $T_{\min}$ was reduced to 188 m. Strong tailwinds encountered on the northeast side of both paths caused V_g to increase, thus increasing $\dot{\chi}^p$ for the path. Consequently, the MAV was not able to match the increase in heading rate required to accurately follow the path in these regions. Conversely, on the other side of the high-frequency path where the MAV encountered headwinds, the MAV was able to more accurately follow the path due to a decrease in V_g . Overall, these results indicate that the MAV was able to

follow various commanded paths in windy conditions with only minor following error.

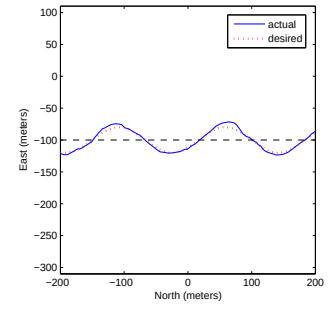
During these flight tests, the MAV used the numerical computation of $\dot{\chi}^p$, given in (21), for path following. However, for comparison purposes $\dot{\chi}^p$ was also computed using the analytical approach in (19). Data comparing these approaches are shown in Figure 7. While the numerical method is noisier than the analytical method, it is easier to implement, requires less computation time, and can be employed for all smooth arbitrary curved paths. If computation time is not an issue, then $\dot{\chi}^p$ should be computed using the explicit solution whenever possible because it is more precise than using numerical derivatives and is less susceptible to noise.

VI. Conclusions

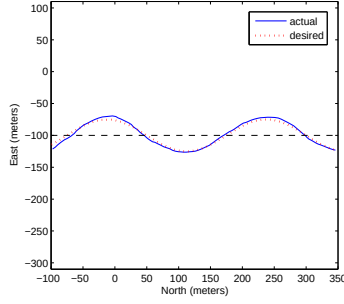
A new method for MAVs to follow curved paths has been introduced. It has been shown using Lyapunov stability criteria that this vector field approach provides asymptotic path following for curved path segments even in the presence of constant wind disturbances. Practical limitations on parameters have been established that provide conditions on the choice of feedback gains. In addition, bounds have been placed on following error using principles of input-to-state stability. The effectiveness of the vector field method has been verified experimentally using fixed-wing MAVs. Lateral-tracking errors for this method averaged 3.4 m for amplitudes ranging between 10 and 100 m and spatial periods between 125 and 500 m. Future work will include flight tests that follow a wider variety of curved paths, not just sinusoids, to further explore the set of flyable paths for MAVs similar to the one used for these tests.



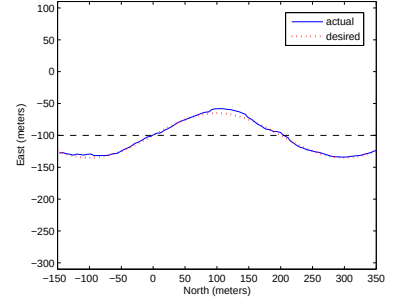
(a) $A = 10$ m, $T = 125$ m, RMSE = 5.1 m



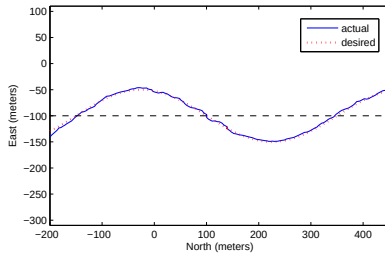
(b) $A = 20$ m, $T = 167$ m, RMSE = 3.3 m



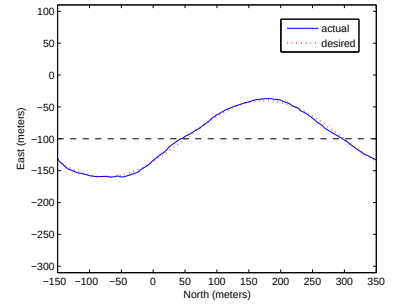
(c) $A = 25$ m, $T = 250$ m, RMSE = 3.2 m



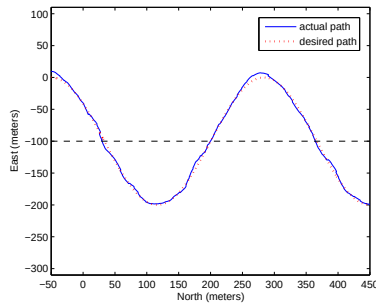
(d) $A = 35$ m, $T = 400$ m, RMSE = 2.9 m



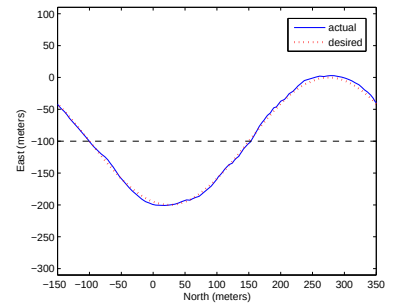
(e) $A = 50$ m, $T = 500$ m, RMSE = 2.5 m



(f) $A = 60$ m, $T = 500$ m, RMSE = 3.0 m



(g) $A = 100$ m, $T = 350$ m, RMSE = 3.7 m



(h) $A = 100$ m, $T = 500$ m, RMSE = 3.4 m

Figure 5. Flight test results for curved path following. The average wind speed was 18 percent of the MAV's airspeed. The desired path is shown in dotted red and the actual path is shown in solid blue.

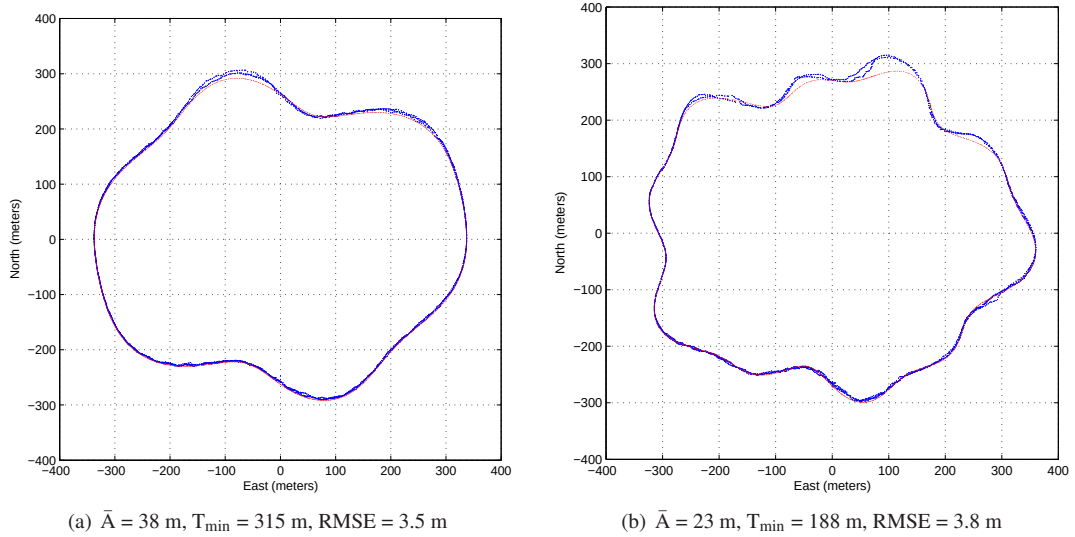


Figure 6. Final flight test results of a MAV following two irregularly shaped circular paths in a counterclockwise direction. The planned path is shown in red while the actual path is shown in blue.

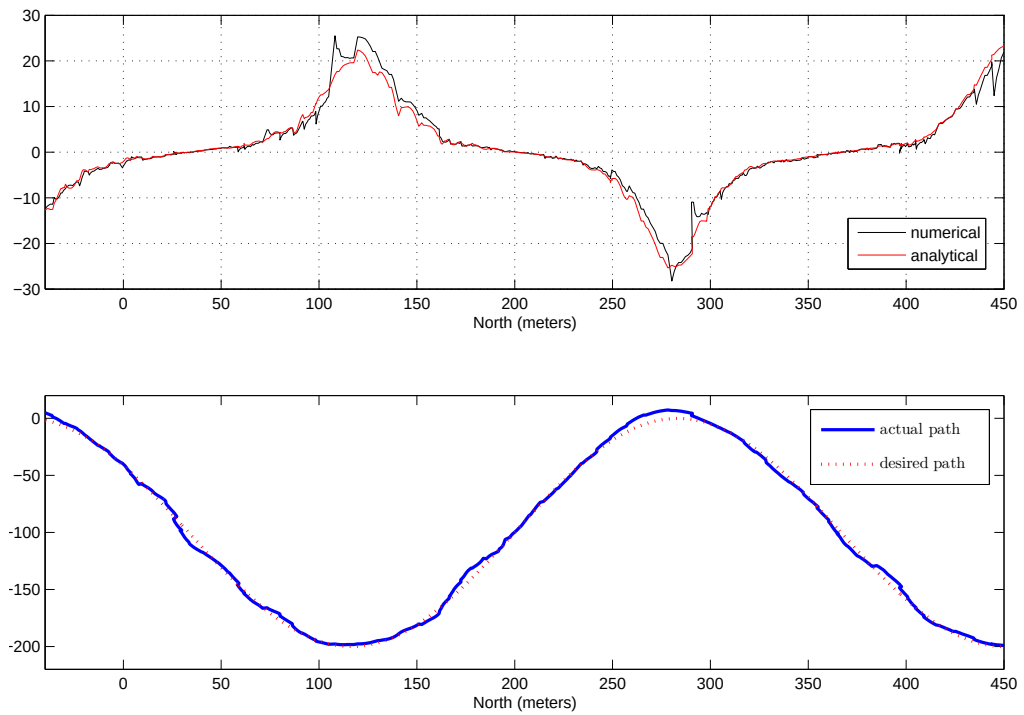


Figure 7. Flight results comparing the numerical and analytical results for $\dot{\chi}^P$. The top figure shows flight data comparing analytical results (red) and numerical results (black) for $\dot{\chi}^P$, in degrees per second, for a MAV following the sinusoidal path shown in the lower figure. In the lower figure the desired path is indicated by the dotted red line and the actual path is indicated by solid blue line. The average wind speed for this flight test was 18 percent of the MAV's airspeed.

Appendix

Proof: It has been shown that the control command in (12) will drive the MAV inside the sliding mode boundary region in finite time. However, to show that the system converges to the origin $(d, \tilde{\chi}) = (0, 0)$ we define the Lyapunov function

$$W = \frac{1}{2}d^2 + \frac{1}{2}\rho\tilde{\chi}^2 \quad (38)$$

where ρ is a positive scale factor that is used to weight the terms so that they are similar in magnitude. For the system to converge $\dot{W}$ must be negative definite. Differentiating (38) we obtain

$$\begin{aligned} \dot{W} &= d\dot{d} + \rho\tilde{\chi}\dot{\tilde{\chi}} \\ &= dV_g \sin(\chi - \chi^p) + \rho\tilde{\chi}\alpha \left(\chi^c - \chi - \frac{\dot{\chi}^p}{\alpha} + \frac{2\chi^\infty}{\alpha\pi} \frac{k}{1 + (kd)^2} V_g \sin(\chi - \chi^p) \right). \end{aligned}$$

Inside the boundary region χ^c can be written as

$$\chi^c = \chi + \frac{\dot{\chi}^p}{\alpha} - \frac{1}{\alpha} \frac{2\chi^\infty}{\mu\pi} \frac{k}{1 + (kd)^2} V_g \sin(\chi - \chi^p) - \frac{\beta}{\alpha} \frac{\tilde{\chi}}{\epsilon}. \quad (39)$$

To simplify the development, we will let

$$\tilde{\chi}^d(d) \triangleq \chi^d(d) - \chi^p = -\frac{2\chi^\infty}{\mu\pi} \tan^{-1}(kd). \quad (40)$$

Substituting (39) and (40) into $\dot{W}$ gives

$$\begin{aligned} \dot{W} &= V_g d \sin(\tilde{\chi}^d(d) + \tilde{\chi}) - \frac{\rho\beta}{\epsilon} \tilde{\chi}^2 \\ &= -\frac{\rho\beta}{\epsilon} \tilde{\chi}^2 + V_g d \sin(\tilde{\chi}^d(d)) + V_g d (\sin(\tilde{\chi}^d(d) + \tilde{\chi}) - \sin(\tilde{\chi}^d(d))) \\ &\leq -\frac{\rho\beta}{\epsilon} \tilde{\chi}^2 + V_g d \sin(\tilde{\chi}^d(d)) + V_g |d| |\sin(\tilde{\chi}^d(d) + \tilde{\chi}) - \sin(\tilde{\chi}^d(d))|. \end{aligned}$$

This expression can be simplified by noting that

$$\begin{aligned} |\sin(\tilde{\chi}^d(d) + \tilde{\chi}) - \sin(\tilde{\chi}^d(d))| &= |\sin(\tilde{\chi}^d(d)) \cos(\tilde{\chi}) + \cos(\tilde{\chi}^d(d)) \sin \tilde{\chi} - \sin(\tilde{\chi}^d(d))| \\ &= |\sin(\tilde{\chi}^d(d)) (\cos(\tilde{\chi}) - 1) + \cos(\tilde{\chi}^d(d)) \sin(\tilde{\chi})| \\ &\leq |\cos(\tilde{\chi}) - 1| + |\sin(\tilde{\chi})| \\ &\leq 2|\tilde{\chi}|, \end{aligned}$$

to yield

$$\begin{aligned} \dot{W} &\leq -\frac{\rho\beta}{\epsilon} \tilde{\chi}^2 + 2V_g |d| |\tilde{\chi}| + V_g d \sin(\tilde{\chi}^d(d)) \\ &\leq -\frac{\rho\beta}{\epsilon} \tilde{\chi}^2 + 2V_g |d| |\tilde{\chi}| - V_g d \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(kd)\right). \end{aligned}$$

Next, let

$$\phi(d) \triangleq d \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(kd)\right).$$

Noting that $\phi(d) \approx \frac{2\chi^\infty k}{\pi} d^2$ for small values of kd and $\phi(d) \approx (\sin \chi^\infty) d$ for large kd . Consider the function

$$\varphi(d) = \begin{cases} \frac{2\chi^\infty k}{\mu\pi} d^2 & \text{if } |d| \leq \bar{d} \\ \frac{2\chi^\infty k \bar{d}}{\mu\pi} (2|d| - \bar{d}) & \text{if } |d| > \bar{d} \end{cases}, \quad (41)$$

where $\bar{d}$ is arbitrary. Our objective is to find μ that ensures that $0 < \varphi(d) \leq \phi(d)$, which will guarantee that $\dot{W} < 0$. To accomplish this, note that both ϕ and φ are symmetric functions in d . Therefore, we will restrict our attention to the

case where $d \geq 0$ without loss of generality. To show that $\varphi(d) \leq \phi(d)$ we will exploit the fact that if two functions f and g satisfy $f(0) = g(0)$ and $f'(d) \leq g'(d)$ for $d \geq 0$, then $f(d) \leq g(d)$ for $d \geq 0$. First, consider the case when $0 \leq d \leq \bar{d}$ where

$$\begin{aligned}\phi'(d) &= \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(kd)\right) \\ &\quad + \frac{2\chi^\infty k}{\pi} d \left[\frac{\cos\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(kd)\right)}{1 + (kd)^2} \right] \\ &\geq \frac{4\chi^\infty k}{\pi} d \left[\frac{1}{2} \frac{\cos\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(kd)\right)}{1 + (kd)^2} \right] \\ &\geq \frac{4\chi^\infty k}{\pi} d \left[\frac{1}{2} \frac{\cos\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)}{1 + (k\bar{d})^2} \right] \\ &\geq \frac{4\chi^\infty k}{\mu\pi} d \\ &= \varphi'(d)\end{aligned}$$

when

$$\mu \geq \frac{2(1 + (k\bar{d})^2)}{\cos\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)}. \quad (42)$$

Second, when $d > \bar{d}$, we have

$$\begin{aligned}\phi(d) &= d \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(kd)\right) \\ &\geq d \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right),\end{aligned}$$

which implies that $\phi(d) \geq \varphi(d)$ if

$$\frac{4\chi^\infty k\bar{d}}{\mu\pi} \leq \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right). \quad (43)$$

Solving (43) for μ yields

$$\mu \geq \frac{4\chi^\infty k\bar{d}}{\pi \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)}. \quad (44)$$

Combining (42) and (44) gives

$$\mu \geq \max \left\{ \frac{2(1 + (k\bar{d})^2)}{\cos\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)}, \frac{4\chi^\infty k\bar{d}}{\pi \sin\left(\frac{2\chi^\infty}{\pi} \tan^{-1}(k\bar{d})\right)} \right\} \quad (45)$$

and $\varphi(d) \leq \phi(d)$. This implies that

$$\dot{W} \leq -\frac{\rho\beta}{\epsilon} \tilde{\chi}^2 + 2V_g |d| |\tilde{\chi}| - V_g \varphi(d).$$

Rewriting this quadratic equation in matrix form gives

$$\dot{W} \leq -V_g \begin{pmatrix} |\tilde{\chi}| & |d| \end{pmatrix} \begin{pmatrix} \frac{\rho\beta}{\epsilon V_g} & -1 \\ -1 & \frac{4k\chi^\infty}{\mu\pi} \end{pmatrix} \begin{pmatrix} |\tilde{\chi}| \\ |d| \end{pmatrix}. \quad (46)$$

The quadratic nature of $\dot{W}$ guarantees asymptotic stability if it is negative definite. By observation $\dot{W}$ is negative definite if

$$\left(\frac{\rho\beta}{\epsilon V_g}\right) \left(\frac{4\chi^\infty k}{\mu\pi}\right) > 1. \quad (47)$$

For $|d| > \bar{d}$, we get

$$\dot{W} \leq -\frac{\rho\beta}{\epsilon} \tilde{\chi}^2 + 2V_g |y| |\tilde{\chi}| - \frac{2V_g \chi^\infty k \bar{d}}{\mu\pi} (2|d| - \bar{d}) \quad (48)$$

$$\leq 2V_g |d| \epsilon - \frac{2V_g \chi^\infty k \bar{d}}{\mu\pi} |d| \quad (49)$$

$$= 2V_g |d| \left(\epsilon - \frac{\chi^\infty k \bar{d}}{\mu\pi} |d| \right), \quad (50)$$

which is less than zero if

$$\frac{\chi^\infty k \bar{d}}{\mu\epsilon\pi} > 1, \quad (51)$$

thus implying asymptotic stability when $|d| > \bar{d}$. ■

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